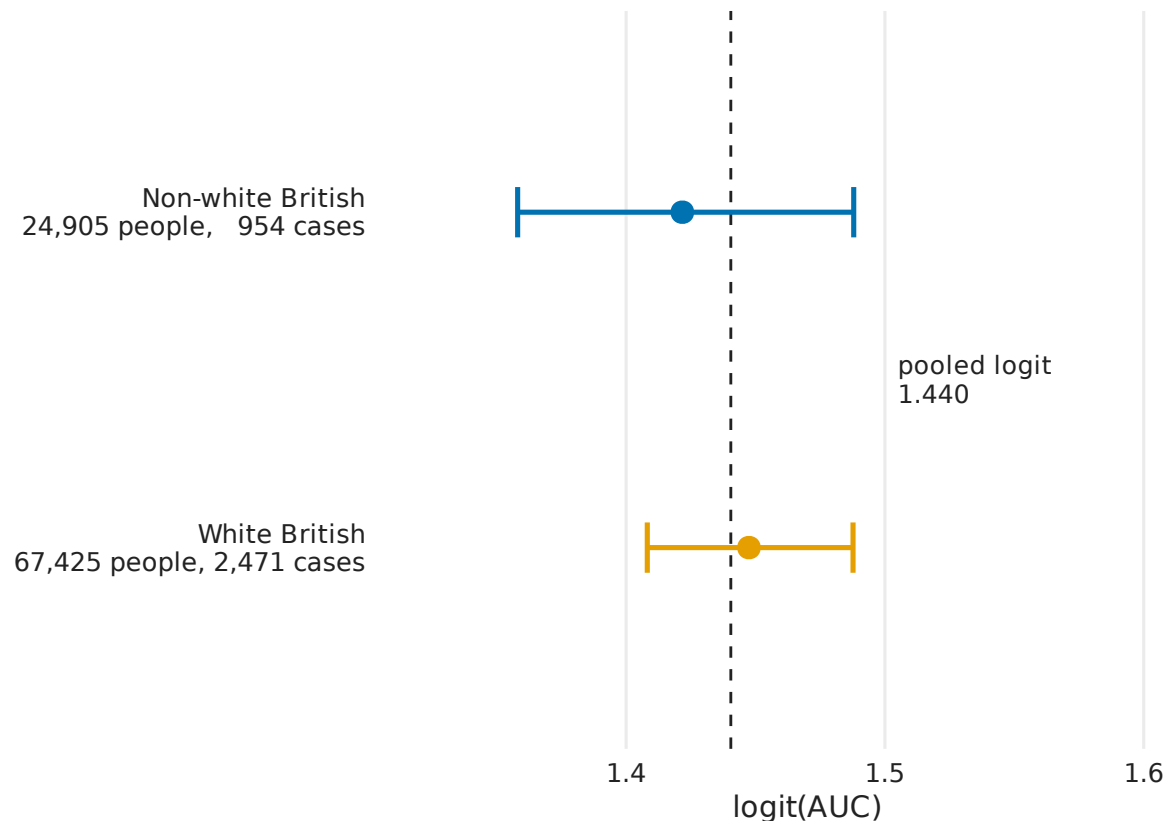


2. Q is computed on the logit of AUC, not on AUC itself

A difference of 0.01 in AUC is not the same precision everywhere between 0 and 1. The logit is the scale on which the weights, the mean, and Q are calculated.

Both studies on the logit scale



Basic functions, one study at a time

Non-white British

$\text{AUC} = 0.80561$

$\text{eta} = \log(\text{AUC} / (1 - \text{AUC})) = 1.422$

logit interval = 1.358 to 1.488

$\text{SE} = (\text{upper} - \text{lower}) / (2 \times 1.96) = 0.0331$

White British

$\text{AUC} = 0.80961$

$\text{eta} = \log(\text{AUC} / (1 - \text{AUC})) = 1.447$

logit interval = 1.408 to 1.488

$\text{SE} = (\text{upper} - \text{lower}) / (2 \times 1.96) = 0.0203$

The larger study has the narrower interval, so its SE is smaller. That is the only reason it will receive more weight.

The dashed line is the weighted mean of the two logits. Mapping it back with `plogis()` returns the pooled AUC in plot 1.